EMILIO GUADARRAMA

ELECTRONICS & SEMICONDUCTOR ENGINEERING STUDENT

EXPECTED GRADUATION · JUNE 2029

Tecnológico de Monterrey

ABOUT ME

Engineering student who designs, builds and flies rocket flight computers, from multilayer PCB design to lab bring-up, flight testing and failure analysis.

EDUCATION

Tecnológico de Monterrey
B.S. in Electronics and Semiconductor Engineering
Aug 2025 – Jun 2029 (expected)

TECHNICAL SKILLS

HARDWARE & PCB
Schematic capture · 4-layer PCB layout
RF layout · Power distribution · BOM management

SOFTWARE & TOOLS
Altium Designer · **Verilog** · KiCad
Python (pandas, PySide6) · Git / GitHub
LTspice · NI Multisim

EMBEDDED
ESP32-S3 · C/C++ · SPI · I²C · UART · USB
LoRa · GNSS · Sensor integration

LAB & TEST
Oscilloscope · Signal generator
Bench power supply · Multimeter
Board bring-up · Failure analysis

ASSEMBLY & MANUFACTURING
DFM for turnkey SMT · Stencil printing
Reflow oven · Soldering & hot-air rework

LANGUAGES

English: C1 (TOEFL iBT 99)
Spanish: Native
German: Elementary

CONTACT

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EXPERIENCE

Ignitia Rocket Lab — Vice President & Avionics / Electronics Lead

Tecnológico de Monterrey **Aug 2025 – Present · Zapopan, Jalisco**

- **Lead hardware designer** of IGNIS, an **open-source rocket flight computer** (ESP32-S3, LoRa telemetry, GNSS, IMU, barometer, two opto-isolated pyro channels). Responsible for architecture, schematics, 4-layer PCB layout, RF and power design, sourcing, bring-up and lab debugging.
- Developed **successive IGNIS revisions (v1 to v5)** based on issues found during bring-up, bench and flight testing, including floating sensor pins, shared-SPI initialization, USB-C net errors, pyro-channel wiring and watchdog resets.
- Flew IGNIS v2.2 as the primary flight computer at **LASC 2026 (Brazil)**. Analyzed test and competition **flight logs in Python**; measured 188 ms barometric apogee-detection latency.
- Redesigned v5 for **turnkey SMT assembly** at JLCPCB (109 placements, 40 BOM lines) after component availability problems on v4. Checked stock and lifecycle before layout and prepared a manufacturing package for **repeatable production** independent of the original designer.
- Defined **firmware requirements** and validated successive flight-software changes through bench and flight testing: 8-state flight state machine, barometric apogee detection, safety interlocks, LoRa telemetry and SD logging.
- **Vice President** of a 30-member team; lead the 9-member avionics & electronics area and train new members on flight-computer design and assembly.

FIRST Robotics Competition — Team Daedalus

Mechanical Lead & Mentor **Jan 2023 – May 2025 · Zapopan, Jalisco**

- **Mechanical Lead & Mentor (2025):** led the team's manufacturing and mentored new members through the Monterrey and León Regionals.
- **Mechanical Lead (2024):** led manufacturing and was a member of the drive team; won the Hermosillo Regional and qualified for the FIRST World Championship in Houston.
- **Robotics Technician (2023):** drive team, scouting and mechanical maintenance.

AWARDS

- **LASC 2026** — 10th of 100 teams, Satellite Challenge (CanSat Mission Leader)
- **FIRST World Championship qualifier** — Hermosillo Regional winner, 2024
- **ENMICE 2025** — 2nd place, project presentation (Wirikuta test bench); Honorary Recognition for Team Conduct
- **Tec de Monterrey Engineering Expo 2025** — 1st place, Physical Prototype (Ignitia Rocket Lab)

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TECHNICAL PORTFOLIO

IGNIS Flight Computer Platform — v1 to v5 (open source)

Lead Hardware Designer · Ignitia Rocket Lab · 2025 – Present

Custom flight computer for student rocketry: senses the flight, detects apogee in real time, fires the recovery charges, streams LoRa telemetry and logs every sample. Each revision was driven by findings from the one before it.

v1	v2	v2.2 · Flew	v3	v4	v5 · 2027
Fabricated; too large for the rocket, redesigned before assembly	First assembled; bring-up found schematic errors	Corrected revision; primary flight computer at LASC 2026	Incremental update; never fabricated	Two units built; halted by part availability and a USB-C net error	Sourcing-first, all-SMD, turnkey; Ø75 mm board

- **Hardware:** 4-layer stackup with dedicated ground reference for the RF microstrip, via-fenced RF keepout, via-in-pad MCU thermal pad, opto-isolated MOSFET pyro drivers with continuity sensing, USB-C with ESD protection.
- **Manufacturing:** v2/v2.2 assembled in-house (laser-cut stencil, reflow, rework). v5 moves to full JLCPCB assembly with standardized 0603/0805 passives, alternate parts for the IMU and barometer, and 37 of 40 BOM lines also stocked by a Mexican distributor for local repair.
- **Mechanical integration:** v5 is a Ø75 mm circular board mounted across the airframe instead of along it, shortening the avionics bay and fitting the team's smallest rocket.
- **Ground segment:** Python/PySide6 ground station with live charts, CSV logging and dual-antenna diversity receivers.

Wrong pressure readings
Barometer read 497 hPa instead of ~830 hPa. **Root cause:** floating chip-select pin. Fixed in v2.2; v5 defines every strap pin.

IMU lost after impact
IMU invalid for 12,970 log rows while the barometer on the same I²C bus kept working. **Root cause:** floating address pin. Address pin defined in the v5 design.

False liftoff in testing
A 40 ms debounce let hand movement trigger "boost". **Fix:** 250 ms debounce plus a 30 m minimum-altitude interlock before any deployment.

Tools: Altium Designer · ESP32-S3 · RFM95W LoRa · u-blox GNSS (GPS) · C/C++ · Python · JLCPCB / LCSC

Open source: github.com/IgnitiaRocketLab/IGNIS-v2-FlightComputer github.com/Emilio-Guadarrama/IGNIS-v5-FlightComputer

LASC 2026 CanSat — Satellite Challenge

Mission Leader · Ignitia Rocket Lab · 2026 · 10th of 100 teams

Led the team's CanSat mission, providing technical direction and supervision. The avionics are a simplified derivative of IGNIS with added sensors that measure the vibration environment affecting the rocket's flight, LoRa telemetry and independent parachute recovery, within a ~300 g, ~$800 USD budget.

Tools: ESP32 · RFM95W LoRa · Environmental and vibration sensors · Python

Smart Irrigation System — Water Growth Solutions

Team project · Sustainable Technology & Renewable Energy

Automated vertical irrigation system with Arduino, sensors, pumps and solar power, aligned with six UN Sustainable Development Goals.

Tools: Arduino · Sensors · Bluetooth / Wi-Fi · Web interface · Solar power

CERTIFICATIONS & HONORS

TOEFL iBT — Score 99 (C1 CEFR)
ETS · April 2025

Advanced Robotics Course
Asociación de Robótica Aplicada y Ciencias de la Tecnología (UNAM) · July 2024 · 45 h · Final grade 10/10

Honorable Mention — Bicultural High School
Tecnológico de Monterrey · May 2025 · **Academic excellence**, GPA 96/100

VOLUNTEERING

TECHO México — Crew Leader
Oct 2024 – Present · Jalisco, Mexico

- **Crew Leader:** Third build, Jauja community, Tonalá
- **Co-leader:** second build, Sergio Barrios Indigenous Community, Tlajomulco
- **Volunteer:** first build, Sergio Barrios Indigenous Community

Built emergency housing alongside families in vulnerable communities.